HANS GUNAWAN

Information Systems Undergraduate | Software and AI Systems Engineer

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EDUCATION

Universitas Kristen Duta Wacana (UKDW), Yogyakarta | B.Sc. Information Systems Sep 2023 - Expected graduation: Aug 2027

- GPA 3.79/4.00. Grade A in Algorithms and Data Structures, Software Engineering, AI for Trading, Decision Support Systems, and Capstone Project. 2nd Place, University-Level Outstanding Student Award (PILMAPRES) 2025.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, Java, Rust, SQL, Solidity

Backend and Systems: Node.js, FastAPI, Fastify, Laravel, REST APIs, relayers and keepers, async workers (Playwright), load testing, unit and invariant testing (Foundry)

Infrastructure: Docker, AWS (EC2), Nginx, GitHub Actions, Git, Railway, Vercel, Cloudflare Workers, PostgreSQL, Redis, Supabase

AI: LLM pipelines (Claude API structured outputs, model fallback), multi-agent workflows, Model Context Protocol (MCP), Langfuse tracing, XGBoost, YOLOv8

PROJECTS

PrintPitt: Real-Time Print-Job Marketplace on Monad Oct 2026

2nd Place, Monad Blitz Jogja Vol. 2 | Team of 3: contracts, backend, and infrastructure

- Built the contract (43 unit tests plus escrow invariants), a Node relayer for gasless orders, a keeper, an encrypted document store (AES-256-GCM), and an Envio indexer; deployed and verified on Monad mainnet.
- Load-tested with 30 concurrent users on testnet (relay p95 10.4 s, 13/13 checks, zero backend failures). Tuned for chain specifics: gas limits measured with `eth_estimateGas`, per-block receipts with `eth_getBlockReceipts`.

Applify: Human-in-the-Loop AI Agent for Job Applications 2026

1st Place, national Solve-It Challenge 2026 | 3rd Place, Udayana Business Plan Competition 2026 | Team of 3, project manager and full-stack developer

- Designed a 4-layer system (React, Supabase, Fastify, asynchronous Playwright workers) with a pause-before-submit approval step, an LLM pipeline with model fallback and Langfuse tracing, deployed on AWS with Docker and Nginx.
- Tested on 189 real vacancies: 614/614 form fields detected, 100/100 submit controls correct, 20/20 row-level-security checks passed. First-author paper presented at ICIC-APTIKOM 2026.

Reeve: Business SOPs Turned Into Audited, Self-Running Code Oct 2026

CreateOS x Stripe Hackathon | Solo

- An LLM turns a plain-language SOP into rules, a compiler generates the Python, and an independent Auditor verifies it with mutation testing (30/30 planted defects caught) before money moves. Jobs run in an isolated microVM with outbound network blocked; every run is written to a hash-chained ledger.

Cult of the Digital Oracle: Autonomous AI Agents With On-Chain Accountability May - Jun 2026

1st Place and $8,500, international Mantle Turing Test Hackathon | Full-stack developer and project manager

- Three AI agents publish daily prophecies and a separate AI judge issues signed (EIP-712) verdicts; a 15,000-entity world runs in a Web Worker at 60 FPS (PixiJS), with full contract test suites.

Fabius, Lencana, and Nusastakes Sep - Oct 2026

- **Fabius** (solo) and **Lencana** (lead builder of 3): both Top 10 of 228 at the Indonesia Web3 Hackathon 2026. Fabius is a Python engine with refusal gates and an MCP server; Lencana issues verifiable W3C credentials and passed the 1EdTech validator with 0 errors.
- **Nusastakes** (solo, Monad Metropolis Hackathon): ERC-4626 vaults with passkey sign-in, a gas relayer, and 229 contract tests passing. **JitsuinFi** (ZECATHON 2026, in development): Rust and WebAssembly invoice-financing registry on Zcash testnet.

EXPERIENCE

Lead Engineer, MBG National Nutrition Program ERP | Synapse Labs (Kampus Merdeka MBKM) Feb - May 2026

- Led 6 engineers to deliver a fully tested, 10-module MVP ERP (Laravel 12, PostgreSQL) with per-kitchen data isolation and FIFO/FEFO stock logic.

Teaching Assistant, Department of Information Systems, UKDW Aug 2025 - Present

- Assist students in Database Design (SQL, PL/SQL) and Database Systems; appointed for Web Frameworks in 2026/2027.

AWARDS AND PUBLICATIONS

- **1st Place**, Solve-It Challenge 2026 (national); **3rd Place**, Udayana Business Plan Competition 2026 (national); **1st Place** ($8,500), Mantle Turing Test Hackathon 2026; **2nd Place**, Monad Blitz Jogja Vol. 2; **Top 10** with two projects, Indonesia Web3 Hackathon 2026; **Top 80 of 300+**, 0G APAC Hackathon 2026.
- **ICIC-APTIKOM 2026:** first author, presented. **IEEE 2025:** co-author, "A Study of Repetition on Virtual Tourism: Understanding the Effect on Memory and Visit Intention".